Kubernetes service:

Already I had created deployment1.yaml and pods are crested for this and now creating clusterip in continuation.

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra (main)
$ cd K8S_Assignment-3

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
Kubernetes_Deployment.pdf  deployment.yaml  deployment1.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f deployment1.yaml
deployment.apps/web-deploy created

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
web-deploy   3/3     3            3           11s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-deploy-86648cb98c-65bkh   1/1     Running   0          18s
web-deploy-86648cb98c-9d8wz   1/1     Running   0          18s
web-deploy-86648cb98c-d5459   1/1     Running   0          18s
```

Creat and apply service clusterip yaml file:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ vim service-clusterip.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
Kubernetes_Deployment.pdf  deployment.yaml  deployment1.yaml  service-clusterip.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ cat service-clusterip.yaml
apiVersion: v1
kind: Service
metadata:
  name: backend
spec:
  selector:
    app: web
  ports:
    - port: 80
      targetPort: 80


hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f service-clusterip.yaml
service/backend configured
```

Verify the service:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get services
NAME       TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
backend    ClusterIP   10.105.23.140   <none>        80/TCP    30m
```

Describe the service:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (m
$ kubectl describe service backend
Name:                     backend
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=web
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.105.23.140
IPs:                      10.105.23.140
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:                10.244.0.31:80,10.244.0.33:80,10.244.0.32:80
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
```

Checking service end points:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get endpoints backend
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS                                      AGE
backend   10.244.0.31:80,10.244.0.32:80,10.244.0.33:80   33m
```

Creating temporary test pod and verifying it:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl run test-pod --image=busybox --restart=Never -- sleep 3600
pod/test-pod created

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pod test-pod
NAME       READY   STATUS    RESTARTS   AGE
test-pod   1/1     Running   0          16s
```

Access the Test Pod:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl exec -it test-pod -- sh
```

Testing clusterip service:

```
/ # wget -O- http://backend
Connecting to backend (10.105.23.140:80)
writing to stdout
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
-                    100% |*****************************************************************
written to stdout
/ # exit
```

Delete temp pod:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra
$ kubectl delete pod test-pod
pod "test-pod" deleted from default namespace
```

Verify all resources:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/web-deploy-86648cb98c-65bkh     1/1     Running   0          73m
pod/web-deploy-86648cb98c-9d8wz     1/1     Running   0          73m
pod/web-deploy-86648cb98c-d5459     1/1     Running   0          73m

NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
service/backend      ClusterIP   10.105.23.140   <none>        80/TCP    85m
service/kubernetes   ClusterIP   10.96.0.1       <none>        443/TCP   8d

NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-deploy   3/3     3            3           73m

NAME                                    DESIRED   CURRENT   READY   AGE
replicaset.apps/web-deploy-86648cb98c   3         3         3       73m
```

Clean up:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl delete -f service-clusterip.yaml
service "backend" deleted from default namespace
```

We already created and applied web-deploy in above case and now continuing nxt steps.

Create and apply servicenode port:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ vim service-nodeport.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
Kubernetes_Deployment.pdf  deployment.yaml  deployment1.yaml  service-clusterip.yaml  service-nodeport.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ cat service-nodeport.yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nodeport
spec:
  type: NodePort
  selector:
    app: web
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080


hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f service-nodeport.yaml
service/web-nodeport created
```

Verify and describe service:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get services
NAME           TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
kubernetes     ClusterIP   10.96.0.1     <none>        443/TCP        8d
web-nodeport   NodePort    10.98.88.45   <none>        80:30080/TCP   3m9s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl describe service web-nodeport
Name:                     web-nodeport
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=web
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.98.88.45
IPs:                      10.98.88.45
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30080/TCP
Endpoints:                10.244.0.32:80,10.244.0.31:80,10.244.0.33:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
```

Check the service endpoints:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get endpoints web-nodeport
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME           ENDPOINTS                                      AGE
web-nodeport   10.244.0.31:80,10.244.0.32:80,10.244.0.33:80   12m
```

Get kuberneties node ip:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get nodes -o wide
NAME       STATUS   ROLES           AGE   VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION                      CONTAINER-RUNTIME
minikube   Ready    control-plane   8d    v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   6.18.33.2-microsoft-standard-WSL2   docker://29.2.1

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ minikube ip
192.168.49.2
```

Running **Minikube with Docker + WSL2**, accessing the Minikube internal IP directly from Windows browser may not work.The minikube service --url command was used to access the NodePort Service from the Windows host because the Kubernetes cluster is running in Minikube using the Docker driver.

Access the Application Using Minikube:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (m
$ minikube service web-nodeport
|-----------|--------------|-------------|---------------------------|
| NAMESPACE |     NAME     | TARGET PORT |            URL            |
|-----------|--------------|-------------|---------------------------|
| default   | web-nodeport | 80          | http://192.168.49.2:30080 |
|-----------|--------------|-------------|---------------------------|
* Starting tunnel for service web-nodeport.
|-----------|--------------|-------------|------------------------|
| NAMESPACE |     NAME     | TARGET PORT |          URL           |
|-----------|--------------|-------------|------------------------|
| default   | web-nodeport |             | http://127.0.0.1:52616 |
|-----------|--------------|-------------|------------------------|
* Opening service default/web-nodeport in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
* Stopping tunnel for service web-nodeport.
```

Verify all resources:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main
$ kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/web-deploy-86648cb98c-65bkh   1/1     Running   0          122m
pod/web-deploy-86648cb98c-9d8wz   1/1     Running   0          122m
pod/web-deploy-86648cb98c-d5459   1/1     Running   0          122m

NAME                   TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
service/kubernetes     ClusterIP   10.96.0.1     <none>        443/TCP        8d
service/web-nodeport   NodePort    10.98.88.45   <none>        80:30080/TCP   38m

NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-deploy   3/3     3            3           122m

NAME                                    DESIRED   CURRENT   READY   AGE
replicaset.apps/web-deploy-86648cb98c   3         3         3       122m
```

Deleting all resources which are created:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl delete service web-nodeport
service "web-nodeport" deleted from default namespace
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_a
$ kubectl delete deployment web-deploy
deployment.apps "web-deploy" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_a
$ ^C

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_a
$ kubectl get pods
No resources found in default namespace.
```